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Section Number: 31841
TAC 249 Final Project Part 2

Part 2

1.

```
db.sunroof.aggregate([
  { $match: { region: "West" } },
  {
    $group: {
      _id: null,
      maxQualified: { $max: "$percent_qualified" },
      avgQualified: { $avg: "$percent_qualified" },
      minQualified: { $min: "$percent_qualified" }
    }
  },
  {
    $project: {
      _id: 0,
      maxQualified: 1,
      avgQualified: 1,
      minQualified: 1
    }
  }
])
```

```
< {
    maxQualified: 96.429936,
    avgQualified: 88.275455,
    minQualified: 75.68344
  }
```

This output shows the highest, average, and lowest percentage of buildings that are suitable for solar installation among states in the West region, giving a quick snapshot of how much suitability varies across the region.

2.

```
db.sunroof.aggregate([
  {
    $group: {
      _id: "$region",
      avg_q: { $avg: "$percent_qualified" }
    }
  },
  {
    $sort: { avg_q: -1 }
  },
  {
    $limit: 1
  }
])
```

```
< {
    _id: 'West',
    avg_q: 88.275455
  }
```

This output shows that the West region has the highest average percentage of buildings suitable for solar installation, with an average of about 88 percent, compared to the other regions.

3.

```
db.sunroof.aggregate([
  {
    $group: {
      _id: null,
      north_sum: { $sum: "$yearly_sunlight_kwh_n" },
      south_sum: { $sum: "$yearly_sunlight_kwh_s" },
      east_sum:  { $sum: "$yearly_sunlight_kwh_e" },
      west_sum:  { $sum: "$yearly_sunlight_kwh_w" },
```

```
    flat_sum:  { $sum: "$yearly_sunlight_kwh_f" }
   }
  },
  {
   $project: {
    _id: 0,
    north_sum: { $divide: ["$north_sum", 10000000] },
    south_sum: { $divide: ["$south_sum", 10000000] },
    east_sum:  { $divide: ["$east_sum", 10000000] },
    west_sum:  { $divide: ["$west_sum", 10000000] },
    flat_sum:  { $divide: ["$flat_sum", 10000000] }
   }
  }
])
```

```
< {
    north_sum: 7407.909472044,
    south_sum: 33028.27,
    east_sum: 19528.24,
    west_sum: 23908.92,
    flat_sum: 63095.5
  }
Enterprise chuang40>
```

This output shows the total yearly sunlight for each roof orientation across all states, scaled down by dividing each total by 10 million for easier comparison.

4.

```
db.sunroof.aggregate([
  { $match: { region: "West" } },
  {
   $project: {
```

```
    _id: 0,
    state_name: 1,
    number_of_panels_total: 1
  }
},
{ $sort: { number_of_panels_total: -1 } },
{ $limit: 3 }
])
```

```
< {
    state_name: 'California',
    number_of_panels_total: 655000000
  }
  {
    state_name: 'Arizona',
    number_of_panels_total: 147000000
  }
  {
    state_name: 'Washington',
    number_of_panels_total: 108000000
  }
```

This output shows the three West region states with the highest total potential number of solar panels, ranked from highest to lowest.

5.

```
db.sunroof.aggregate([
  { $match: { existing_installs_count: { $gt: 1000 } } },
  {
   $group: {
    _id: "$region",
    stateCount: { $sum: 1 }
   }
  }
])
```

```
< {
    _id: 'West',
    stateCount: 9
  }
  {
    _id: 'Midwest',
    stateCount: 6
  }
  {
    _id: 'South',
    stateCount: 9
  }
  {
    _id: 'Northeast',
    stateCount: 5
  }
```

This output shows how many states in each region have more than 1,000 buildings with solar installations, highlighting where solar adoption is more widespread across states.

6.

```
db.sunroof.aggregate([
  {
   $group: {
     _id: "$region",
     totalInstalls: { $sum: "$existing_installs_count" },
     totalQualified: { $sum: "$count_qualified" }
   }
  },
  {
   $project: {
     installRatio: { $divide: ["$totalInstalls", "$totalQualified"] }
   }
  },
  { $sort: { installRatio: 1 } },
  { $limit: 1 }
])
```

```
< {
    _id: 'Midwest',
    installRatio: 0.0013976612468469428
  }
```

This output shows the region with the lowest overall install-to-qualified ratio, calculated using total installs divided by total suitable buildings across the whole region.

7.

```
db.sunroof.aggregate([
  { $match: { region: "Northeast" } },
  {
   $project: {
     _id: 0,
     state_name: 1,
     region: 1,
     carbon_offset_metric_tons: 1
   }
  },
  { $sort: { carbon_offset_metric_tons: -1 } },
  { $limit: 3 }
```

])

```
< {
    state_name: 'Pennsylvania',
    region: 'Northeast',
    carbon_offset_metric_tons: 25700000
  }
  {
    state_name: 'New York',
    region: 'Northeast',
    carbon_offset_metric_tons: 21900000
  }
  {
    state_name: 'New Jersey',
    region: 'Northeast',
    carbon_offset_metric_tons: 21400000
  }
```

This output shows the three Northeastern states with the highest total carbon offset, ranked from highest to lowest, indicating where solar installations have the greatest environmental impact in the region.

8.

```
db.sunroof.aggregate([
  {
    $project: {
      initial: { $substr: ["$state_name", 0, 1] }
    }
  },
  {
    $group: {
```

```
    _id: "$initial",
    stateCount: { $sum: 1 }
  }
},
{
  $match: {
    stateCount: { $gte: 4 }
  }
}
])
```

```
< {
    _id: 'I',
    stateCount: 4
  }
  {
    _id: 'M',
    stateCount: 8
  }
  {
    _id: 'W',
    stateCount: 4
  }
  {
    _id: 'N',
    stateCount: 8
  }
  {
    _id: 'A',
    stateCount: 4
  }
```

This output groups states by the first letter of their name and shows how many states start with each letter, keeping only letters that appear at least four times.